Genome size as a prediction of plant adaptation across environments

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Abstract

Genome size is an important factor of evolutionary since it relies on how chromosomes will eventually fold and express, and then to its genotypes and phenotypes. It has been debated with multiple hypotheses to explain genome size variation, from the influence from evolutionary history to environments and adaptation.
In this assignment, I examined genome size using lars2 to investigate factors that contribute to its variation and if they harbor linear relationship to genome size or not.

1 Introduction

Genome size has been an important factor of genome. Angiosperm species exhibit a wide range of variation in genome size that many hypotheses have been proposed to explain the variation.

2 Methods

We use lars2 package from caret to examine relationship between genome size and other factors.

We proposed a model of genome size, that genome size is explainable through range size and bioclimatic variables

$$\log_{10}(\text{Genome size}) \sim \log_{10}(\text{Range size}) + \text{number of chromosome} + \text{Bioclimatic variables}$$

Here, Genome size is the genome size of species recorded in this dataset and the response variable that we are interested in. Range size is the range size of species recorded, number of chromosome is the number of chromosome of that speices, and Bioclimatic variables are reported in SI.

See SI section 1 for more details on bioclimatic variables.

3 Results

On the training set, lars2 had an RMSE of 0.453 and an R^2 of 0.247. Then, we tunned the trained model with more steps at it did not improve the performance with RMSE of 0.455 and an R^2 of 0.019. Hence, we used intial model to test on test set. On the test set, it had an RMSE of 0.462 and an R^2 of 0.233.

The tuning graph of tuning process is reported here:

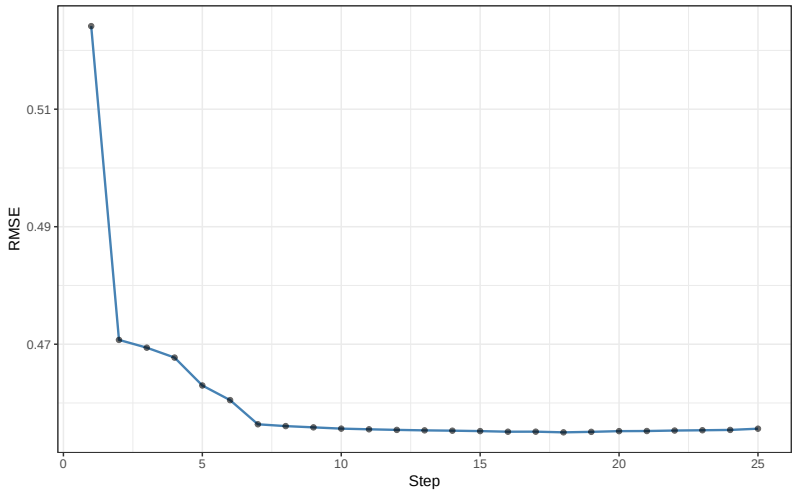


Figure 1: Tuning steps of lars2 and corresponding RMSE of each step